BRAF fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes BRAF gene fusions from the msk_impact_50k_2026 study, covering 179 fusion events. 151/179 fusions (84.4%) were in-frame. 163/179 fusions (91.1%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test (p=0.0133676, statistically significant at alpha=0.05). Compared to the literature benchmark (PMC5461196: 100.0% in-frame, 100.0% domain-retained), this run observed 84.4% in-frame and 91.1% domain retention.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 179 analyzed fusion events, identifying 67 distinct fusion partner genes. The most frequent partner was KIAA1549, observed in 43 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions retained the domain, p=0.0133676 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00699301.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 142/151 (94.0%) of in-frame fusions disrupted the domain, p=0.041705 (statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.963504.

Cutpoint detection

Cutpoint detection scanned 178 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 380 aa (permutation-corrected p=0.001998, statistically significant at alpha=0.05). This is -78 aa from the nearest configured domain boundary at 458 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=163) and "not_retained" (n=15). For Frame_status, retained: 142/152 (93.4%, 95% CI 88.3%-96.4%); not_retained: 9/11 (81.8%, 95% CI 52.3%-94.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 52.0736 vs 36.4, p=0.11965 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score produced results for this run (67 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: BRAF has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
KIAA1549	43	0.24022346368715083	0.21553360374650618	0.001614663595055688	0.8847211802950737	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.33366411984501465	1
CUL1	5	0.027932960893854747	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28565588873695824	2
FAM131B	3	0.01675977653631285	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28230393342969573	3
SND1	17	0.09497206703910614	0.21553360374650618	0.001614663595055688	0.8317520556609741	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.28214333215548626	4
AKAP9	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2806279557760644	5
CARMIL1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2806279557760644	6
EXOC4	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2806279557760644	7
MAD1L1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2806279557760644	8
ABCC1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CAPZA2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	10
CCDC6	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	11
CMAS	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	12
ERC1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	13
KLHL12	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	14
LRBA	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	15
MINDY4	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	16
OSBPL7	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	17
PHTF2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	18
PRIM2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	19

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PRKAR1B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	20
PTPRZ1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	21
RGS3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	22
SNX8	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	23
TRA2B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.989247311827957	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2789519781224331	24
TRIM24	8	0.0446927374301676	0.21553360374650618	0.001614663595055688	0.8951612903225806	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.27657091847204573	25
NRF1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.9247311827956989	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2709505364212257	26
AGAP3	6	0.0335195530726257	0.21553360374650618	0.001614663595055688	0.8637992831541219	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26851466208951436	27
GTF2IRD1	3	0.01675977653631285	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.26294909472001826	28
ATF7	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	29

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CLEC2L	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	30
DBI	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	31
DOCK4	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	32
GIPC2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	33
GTF2I	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	34
KCND2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	35
NFIA	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	36
OSBPL9	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	37
PAK2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	38
PRKAR2B	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	39

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RRBP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	40
SETBP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	41
SLC45A3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	42
SPOCK1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	43
ZC3HAV1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	44
ZNF207	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.8602150537634409	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.25959713941275575	45
CDK5RAP2	4	0.0223463687150838	0.21553360374650618	0.001614663595055688	0.803763440860215	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2561573304381657	46
CLIP1	2	0.0111731843575419	0.21553360374650618	0.001614663595055688	0.7096774193548387	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2386924719050967	47
AGK	14	0.0782122905027933	0.21553360374650618	0.001614663595055688	0.41167434715821816	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21410374291917902	48
MKRN1	10	0.055865921787709494	0.21553360374650618	0.001614663595055688	0.4354838709677419	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.21097126087608245	49

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TMPRSS2	4	0.0223463687150838	0.21553360374650618	0.001614663595055688	0.4408602150537635	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20172184656719797	50
ABCC2	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	51
ITGB5	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	52
MKLN1	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	53
NEO1	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	54
PWWP2A	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	55
TBXAS1	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	56
UBE2H	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	57
WEE2	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.4301075268817204	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19508101038049763	58
PJA2	1	0.0055865921787709 5	0.21553360374650618	0.001614663595055688	0.3763440860215054	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1870164942514654	59

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
SVOPL	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3763440860215054	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1870164942514654	60
EPB41	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18540359102565895	61
PLPP1	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18540359102565895	62
RBM33	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18540359102565895	63
RBMS3	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18540359102565895	64
SCRIB	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18540359102565895	65
TRIM33	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.3655913978494624	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18540359102565895	66
LUC7L2	1	0.00558659217877095	0.21553360374650618	0.001614663595055688	0.0	0.7468257003897063	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1305648813482396	67

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
238	1	162	0	15	None	1.0	-0.0
287	1	162	1	14	0.08641975308641975	0.16187392877547135	0.7908230925727642
303	2	161	1	14	0.17391304347826086	0.23330512802756184	0.6320757153442778

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
327	19	144	6	9	0.19791666666666666	0.00883450422715351	2.053817817119616
380	19	144	9	6	0.08796296296296297	4.496509404813937e-05	4.347124493887456
381	114	49	9	6	1.5510204081632653	0.5596655980047007	0.2520713878510084
393	140	23	12	3	1.5217391304347827	0.46313499579777323	0.33429240123968906
400	141	22	12	3	1.6022727272727273	0.44662922460047316	0.3500528628421283
438	141	22	14	1	0.4577922077922078	0.6965277300948016	0.15706158880581256
439	163	0	14	1	None	0.08426966292134831	1.0743287432532127

domain_disruption tables

frame_domain_contingency_table

142	22
9	5

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ras_binding	RAS-binding domain	178	14	0	164	0.0
cysteine_rich	Cysteine-rich domain	178	14	1	163	0.005699893955461294

domain_retention tables

frame_domain_contingency_table

142	21
9	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	178	163	0	15	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	163	179	0.9106145251396648

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ABCC1	1
ABCC2	1
AGAP3	6
AGK	14
AKAP9	2
ATF7	1
CAPZA2	1
CARMIL1	2
CCDC6	1
CDK5RAP2	4
CLEC2L	1
CLIP1	2
CMAS	1
CUL1	5
DBI	1
DOCK4	1
EPB41	1
ERC1	1
EXOC4	2
FAM131B	3
GIPC2	1
GTF2I	1
GTF2IRD1	3
ITGB5	1
KCND2	1
KIAA1549	43
KLHL12	1
LRBA	1
LUC7L2	1

Partner_gene	Event_count
MAD1L1	2
MINDY4	1
MKLN1	1
MKRN1	10
NEO1	1
NFIA	1
NRF1	2
OSBPL7	1
OSBPL9	1
PAK2	1
PHTF2	1
PJA2	1
PLPP1	1
PRIM2	1
PRKAR1B	1
PRKAR2B	1
PTPRZ1	1
PWWP2A	1
RBM33	1
RBMS3	1
RGS3	1
RRBP1	1
SCRIB	1
SETBP1	1
SLC45A3	1
SND1	17
SNX8	1
SPOCK1	1
SVOPL	1
TBXAS1	1
TMPRSS2	4
TRA2B	1
TRIM24	8
TRIM33	1
UBE2H	1

Partner_gene	Event_count
WEE2	1
ZC3HAV1	1
ZNF207	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021458-T01-IM6-1	P-0021458-T01-IM6		ABCC1-BRAF	ABCC1	in-frame	three_prime	140493152	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC1 (NM_004996) - BRAF (NM_004333) rearrangement: t(7,16)(q34, p13.11)(chr7:g.140493152::chr16:g.16116105) | Protein Fusion: in frame {ABCC1:BRAF}	NA
EVT-P-0034621-T01-IM6-2	P-0034621-T01-IM6		ABCC2-BRAF	ABCC2	unknown	three_prime	140495500	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ABCC2 (NM_000392) - BRAF (NM_004333) rearrangement: t(7;10)(q34;24.2)(chr7:g.140495500::chr10:g.101559038) | Protein Fusion: mid-exon {ABCC2:BRAF}	NA
EVT-P-0052582-T01-IM6-3	P-0052582-T01-IM6		AGAP3-BRAF	AGAP3	unknown	three_prime	140485658	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1637:AGAP3_c.1177+1690:BRAFinv | Protein Fusion: mid-exon {AGAP3:BRAF}	NA
EVT-P-0050574-T02-IM6-4	P-0050574-T02-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140493208	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1222-2059:AGAP3_c.1140+900:BRAFinv | Protein Fusion: in frame {AGAP3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0039799-T01-IM6-5	P-0039799-T01-IM6		AGAP3-BRAF	AGAP3	in-frame	three_prime	140482317	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) fusion: c.1221+416:AGAP3_c.1314+504:BRAFinv | Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0050931-T01-IM6-6	P-0050931-T01-IM6		AGK-BRAF	AGK	in-frame	three_prime	140499251	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-14294:AGK_c.980+911:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T03-IM6-7	P-0042835-T03-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0042835-T02-IM6-8	P-0042835-T02-IM6		AGK-BRAF	AGK	in-frame	three_prime	140497995	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.102-13644:AGK_c.980+2167:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0050635-T01-IM6-9	P-0050635-T01-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140487501	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.6613-1489:AKAP9_c.1141-117:BRAFinv | Protein Fusion: in frame {AKAP9:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030237-T03-IM6-10	P-0030237-T03-IM6		AKAP9-BRAF	AKAP9	in-frame	three_prime	140492357	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AKAP9 (NM_005751) - BRAF (NM_004333) fusion: c.3318+1378:AKAP9_c.1140+1751:BRAFinv | Protein Fusion: in frame {AKAP9:BRAF}	NA
EVT-P-0023821-T01-IM6-12	P-0023821-T01-IM6		ATF7-BRAF	ATF7	in-frame	three_prime	140483233	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ATF7 (NM_006856) - BRAF (NM_004333) Rearrangement : t(7;12)(q34;q13.13)(chr7:g.140483233::chr12:g. 53913299) | Protein Fusion: in frame {ATF7:BRAF}	NA
EVT-P-0007861-T01-IM5-13	P-0007861-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140490697	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1326+220:AGAP3_c.1141-3313:BRAFinv | Protein fusion: in frame (AGAP3-BRAF)	NA
EVT-P-0021519-T02-IM6-14	P-0021519-T02-IM6		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493112	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) Rearrangement : c.1326+142:AGAP3_c.1140+996:BRAFinv | Protein Fusion: in frame {AGAP3:BRAF}	NA
EVT-P-0009757-T01-IM5-15	P-0009757-T01-IM5		BRAF-AGAP3	AGAP3	in-frame	three_prime	140493871	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGAP3 (NM_031946) - BRAF (NM_004333) rearrangement: c.1222-2154:AGAP3_c.1140+237:BRAFinv | Protein fusion: in frame (AGAP3-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036886-T01-IM6-16	P-0036886-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494309	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+10859:AGK_c.981-42:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0025561-T03-IM6-17	P-0025561-T03-IM6		BRAF-AGK	AGK	in-frame	three_prime	140498719	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 8-18): c.980+1288:BRAF_c.101+772:AGKinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0025561-T01-IM6-18	P-0025561-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498874	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) Rearrangement: c.980+1288:BRAF_c.101+772:AGKinv | Protein Fusion: in frame {BRAF:AGK}	NA
EVT-P-0024227-T01-IM6-19	P-0024227-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140495070	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement: c.101+16796:AGK_c.981-803:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0004477-T01-IM5-20	P-0004477-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140495405	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1 to 2 and BRAF exons 8 to 18): c.101+10076:AGK_c.981-1138:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0023378-T01-IM6-21	P-0023378-T01-IM6		BRAF-AGK	AGK	in-frame	five_prime	140498362	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - AGK (NM_018238) rearrangement: c.980+1800:BRAF_c.101+14574:AGKinv | Protein Fusion: in frame {BRAF:AGK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0001832-T02-IM6-22	P-0001832-T02-IM6		BRAF-AGK	AGK	in-frame	three_prime	140494489	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exon 1 fused in-frame to BRAF exons 8-18): c.102-572:AGK_c.981-222:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0000913-T02-IM5-23	P-0000913-T02-IM5		BRAF-AGK	AGK	in-frame	three_prime	140496386	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) Fusion (AGK exon 2 fused with BRAF exon 8) : c.101+13932:AGK_c.981-2119:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA
EVT-P-0029531-T01-IM6-24	P-0029531-T01-IM6		BRAF-AGK	AGK	unknown	three_prime	140500233	7	303	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion (AGK exons 1-2 fused to BRAF exons 7-18): c.101+4572:AGK_c.909:BRAFinv | Protein Fusion: mid-exon {AGK:BRAF}	NA
EVT-P-0030746-T01-IM6-25	P-0030746-T01-IM6		BRAF-AGK	AGK	in-frame	three_prime	140497533	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) fusion: c.101+8348:AGK_c.980+2629:BRAFinv | Protein Fusion: in frame {AGK:BRAF}	NA
EVT-P-0010945-T01-IM5-26	P-0010945-T01-IM5		BRAF-AGK	AGK	in-frame	three_prime	140497610	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		AGK (NM_018238) - BRAF (NM_004333) rearrangement : c.101+12646:AGK_c.980+2552:BRAFinv | Protein fusion: in frame (AGK-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0049035-T01-IM6-85	P-0049035-T01-IM6		BRAF-CLEC2L	CLEC2L	in-frame	five_prime	140485799	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CLEC2L (NM_001080511) rearrangement: c.1177+1549:BRAF_c.191-5932:CLEC2Linv | Protein Fusion: in frame {BRAF:CLEC2L}	NA
EVT-P-0004795-T01-IM5-87	P-0004795-T01-IM5		BRAF-CUL1	CUL1	in-frame	three_prime	140493811	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1(NM_003592) - BRAF (NM_004333) Rearrangement :c.790-427:CUL1_c.1140+297:BRAFinv | Protein fusion: in frame (CUL1-BRAF)	NA
EVT-P-0022257-T01-IM6-88	P-0022257-T01-IM6		BRAF-EPB41	EPB41	in-frame	three_prime	140482687	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EPB41 (NM_001166005) - BRAF (NM_004333) fusion (EPB41 exons 1-5 fused in-frame with BRAF exons 11-18): t(1;7)(p35.4;q34)(chr1:g.29424227::chr7:g.140482687) | Protein Fusion: in frame {EPB41:BRAF}	NA
EVT-P-0001988-T01-IM3-89	P-0001988-T01-IM3		BRAF-FAM131B	FAM131B	in-frame	three_prime	140489629	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) Deletion : c.175-103:FAM131B_c.1141-2245:BRAFdel | Protein fusion: in frame (FAM131B-BRAF)	NA
EVT-P-0010494-T01-IM5-90	P-0010494-T01-IM5		BRAF-GIPC2	GIPC2	in-frame	three_prime	140487276	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GIPC2 (NM_017655) -BRAF (NM_004333) Rearrangement : t(1;7)(p31.1;q34)(chr1:g.78571528::chr7:g.140487276) | Protein fusion: in frame (GIPC2-BRAF)	NA
EVT-P-0054046-T01-IM6-93	P-0054046-T01-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140484706	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-1749:BRAF_c.2320+10446:GTF2IRD1inv | Protein Fusion: in frame {BRAF:GTF2IRD1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0000828-T03-IM6-94	P-0000828-T03-IM6		BRAF-GTF2IRD1	GTF2IRD1	in-frame	five_prime	140483554	9	393	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - GTF2IRD1 (NM_005685) rearrangement: c.1178-597:BRAF_c.2320+14181:GTF2IRD1inv | Protein Fusion: in frame {BRAF:GTF2IRD1}	NA
EVT-P-0015847-T02-IM6-95	P-0015847-T02-IM6		BRAF-ITGB5	ITGB5	out-of-frame	five_prime	140497019	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - ITGB5 (NM_002213) Rearrangement : t(3,7)(q21.2,q34)(chr3:g.124579522::chr7:g.140497019) | Protein Fusion: out of frame {BRAF:ITGB5}	NA
EVT-P-0014377-T01-IM6-99	P-0014377-T01-IM6		BRAF-NFIA	NFIA	in-frame	three_prime	140486673	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NFIA (NM_001145512) - BRAF (NM_004333) rearrangement: t(1;7)(p31.3;q34)(chr1:g.61836194::chr7:g.140486673) | Protein fusion: in frame (NFIA-BRAF)	NA
EVT-P-0010398-T01-IM5-100	P-0010398-T01-IM5		BRAF-OSBPL9	OSBPL9	in-frame	three_prime	140484912	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL9 (NM_148909) - BRAF (NM_004333) Rearrangement : t(1;7)(p32.3;q34)(chr1:g.52228623::chr7:g.140484912) | Protein fusion: in frame (OSBPL9-BRAF)	NA
EVT-P-0003502-T01-IM5-101	P-0003502-T01-IM5		BRAF-PJA2	PJA2	in-frame	five_prime	140482379	10	438	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		NA | Protein fusion: in frame (BRAF-PJA2)	NA
EVT-P-0036870-T01-IM6-102	P-0036870-T01-IM6		BRAF-PRIM2	PRIM2	in-frame	three_prime	140493232	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRIM2 (NM_000947) - BRAF (NM_004333) Rearrangement : t(6;7)(p11.1;q34)(chr6:g.57410815::chr7:g.140493232) | Protein Fusion: in frame {PRIM2:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009246-T01-IM5-103	P-0009246-T01-IM5		BRAF-RBM33	RBM33	in-frame	three_prime	140482661	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBM33 (NM_053043) - BRAF (NM_004333) rearrangement: c.171+526:RBM33_c.1314+160:BRAFinv | Protein fusion: in frame (RBM33-BRAF)	NA
EVT-P-0016313-T01-IM6-104	P-0016313-T01-IM6		BRAF-SLC45A3	SLC45A3	unknown	three_prime	140486608	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SLC45A3 (NM_033102) - BRAF (NM_004333) fusion (SLC45A3 exons 1 fused to BRAF exons 10-18): t(1;7)(q32.1;q34)(chr1:g.205638452::chr7:g.140486608) | Transcript Fusion {SLC45A3:BRAF}	NA
EVT-P-0064764-T01-IM7-105	P-0064764-T01-IM7		BRAF-SND1	SND1	in-frame	five_prime	140493289	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - SND1 (NM_014390) fusion: c.1140+819:BRAF_c.1039-3950:SND1inv | Protein Fusion: in frame {BRAF:SND1}	NA
EVT-P-0029981-T01-IM6-106	P-0029981-T01-IM6		BRAF-SPOCK1	SPOCK1	in-frame	three_prime	140483862	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SPOCK1 (NM_004598) - BRAF (NM_004333) fusion: t(5;7)(q31.2;q34)(chr5:g.136401840::chr7:g.140483862) | Protein Fusion: in frame {SPOCK1:BRAF}	NA
EVT-P-0064223-T01-IM7-107	P-0064223-T01-IM7		BRAF-TBXAS1	TBXAS1	out-of-frame	five_prime	140494512	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - TBXAS1 (NM_001061) rearrangement: c.981-245:BRAF_c.336+2985:TBXAS1inv | Protein Fusion: out of frame {BRAF:TBXAS1}	NA
EVT-P-0021469-T01-IM6-108	P-0021469-T01-IM6		BRAF-TRIM33	TRIM33	in-frame	three_prime	140481644	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM33 (NM_015906) - BRAF(NM_004333) rearrangement: t(1;7)(p13.2;q34)(chr1:g.114963985::chr7:g.140481644) | Protein Fusion: in frame {TRIM33:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0042835-T03-IM6-109	P-0042835-T03-IM6		BRAF-WEE2	WEE2	unknown	five_prime	140498912	7	327	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - WEE2 (NM_001105558) rearrangement: c.980+1250:BRAF_c.1305:WEE2inv | Protein Fusion: mid-exon {BRAF:WEE2}	NA
EVT-P-0015985-T03-IM6-111	P-0015985-T03-IM6		CAPZA2-BRAF	CAPZA2	in-frame	three_prime	140491425	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CAPZA2 (NM_006136) - BRAF (NM_004333) fusion: c.40-938:CAPZA2_c.1140+2683:BRAFinv | Protein Fusion: in frame {CAPZA2:BRAF}	NA
EVT-P-0033764-T04-IM6-112	P-0033764-T04-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491509	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion: t(6;7)(p22.2;q34)(chr6:g.25554753::chr7:g.140491509) | Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0030796-T01-IM6-113	P-0030796-T01-IM6		CARMIL1-BRAF	CARMIL1	in-frame	three_prime	140491645	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRRC16A (NM_017640) - BRAF (NM_004333) fusion (LRRC16A exons 1-29 fused to BRAF exons 9-18): t(6;7)(p22.2;q34)(chr6:g.25580104::chr7:g.140491645) | Protein Fusion: in frame {LRRC16A:BRAF}	NA
EVT-P-0005834-T01-IM5-114	P-0005834-T01-IM5		CCDC6-BRAF	CCDC6	in-frame	three_prime	140488401	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CCDC6 (NM_005436) -BRAF (NM_004333) Rearrangement : t(10;7)(chr10q21.2;q34)(chr10:g.61627617::chr7:g.140488401) | Protein fusion: in frame (CCDC6-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038194-T01-IM6-115	P-0038194-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140484954	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(q34;q33.2)(chr7:g.140484954::chr9:g.123187156) | Protein Fusion: in frame {CDK5RAP2:BRAF}	NA
EVT-P-0004203-T01-IM5-116	P-0004203-T01-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140491585	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement: t(7;9)(7q34;9q33.2)(chr7:140491585::chr9:123251876) | Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0005461-T02-IM5-117	P-0005461-T02-IM5		CDK5RAP2-BRAF	CDK5RAP2	in-frame	three_prime	140481684	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CDK5RAP2 (NM_018249) - BRAF (NM_004333) rearrangement : t(7;9)(7q34;9q33.2)(chr7:140481684::chr9:123255452) | Protein fusion: in frame (CDK5RAP2-BRAF)	NA
EVT-P-0014969-T01-IM6-118	P-0014969-T01-IM6		CDK5RAP2-BRAF	CDK5RAP2	in-frame	five_prime	140493700	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) - CDK5RAP2 (NM_018249) Rearrangement : t(7;9)(q34;q33.3)(chr7:g.140493700::chr9:g.123269531) | Protein fusion: in frame (BRAF-CDK5RAP2)	NA
EVT-P-0040461-T01-IM6-119	P-0040461-T01-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140488218	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140488218::chr12:g.122833264) | Protein Fusion: in frame {CLIP1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0007266-T02-IM6-120	P-0007266-T02-IM6		CLIP1-BRAF	CLIP1	in-frame	three_prime	140495673	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CLIP1 (NM_001247997) - BRAF (NM_004333) fusion: t(7;12)(q34;q24.31)(chr7:g.140495673::chr12:g.122767026) | Protein Fusion: in frame {CLIP1:BRAF}	NA
EVT-P-0024995-T01-IM6-121	P-0024995-T01-IM6		CMAS-BRAF	CMAS	unknown	three_prime	140488046	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) rearrangement: t(7;12)(q34;p12.1)(chr7:g.140488046::chr12:g.22218755) | Transcript Fusion {CMAS:BRAF}	NA
EVT-P-0039648-T01-IM6-122	P-0039648-T01-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0039648-T03-IM6-123	P-0039648-T03-IM6		CUL1-BRAF	CUL1	in-frame	three_prime	140492893	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+2950:CUL1_c.1140+1215:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0064088-T01-IM7-124	P-0064088-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140489451	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+1956:CUL1_c.1141-2067:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068173-T01-IM7-125	P-0068173-T01-IM7		CUL1-BRAF	CUL1	in-frame	three_prime	140491990	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		CUL1 (NM_003592) - BRAF (NM_004333) fusion: c.789+191:CUL1_c.1140+2118:BRAFinv | Protein Fusion: in frame {CUL1:BRAF}	NA
EVT-P-0050196-T01-IM6-126	P-0050196-T01-IM6		DBI-BRAF	DBI	in-frame	three_prime	140486284	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DBI (NM_001178017) - BRAF (NM_004333) fusion: t(2;7)(q14.2;q34)(chr2:g.120128894::chr7:g.140486284) | Protein Fusion: in frame {DBI:BRAF}	NA
EVT-P-0064588-T01-IM7-127	P-0064588-T01-IM7		DOCK4-BRAF	DOCK4	out-of-frame	three_prime	140483872	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		DOCK4 (NM_014705) - BRAF (NM_004333) fusion: c.4650+298:DOCK4_c.1178-915:BRAFdup | Protein Fusion: out of frame {DOCK4:BRAF}	NA
EVT-P-0014833-T02-IM6-128	P-0014833-T02-IM6		ERC1-BRAF	ERC1	in-frame	three_prime	140490668	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ERC1 (NM_178040) - BRAF (NM_004333) fusion: t(7;12)(q34;p13.33)(chr7:g.140490668::chr12:g.1246645) | Protein Fusion: in frame {ERC1:BRAF}	NA
EVT-P-0036937-T01-IM6-129	P-0036937-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140490241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) fusion: c.1514+93469:EXOC4_c.1141-2857:BRAFinv | Protein Fusion: out of frame {EXOC4:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024206-T01-IM6-130	P-0024206-T01-IM6		EXOC4-BRAF	EXOC4	out-of-frame	three_prime	140487956	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		EXOC4 (NM_021807) - BRAF (NM_004333) rearrangement: c.87-8072:EXOC4_c.1141-572:BRAFinv | Protein Fusion: out of frame {EXOC4:BRAF}	NA
EVT-P-0047918-T04-IM7-131	P-0047918-T04-IM7		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel | Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0047918-T02-IM6-132	P-0047918-T02-IM6		FAM131B-BRAF	FAM131B	in-frame	three_prime	140487437	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM131B (NM_001031690) - BRAF (NM_004333) fusion: c.175-20:FAM131B_c.1141-53:BRAFdel | Protein Fusion: in frame {FAM131B:BRAF}	NA
EVT-P-0044845-T01-IM6-134	P-0044845-T01-IM6		GTF2I-BRAF	GTF2I	unknown	three_prime	140485631	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2I (NM_032999) - BRAF (NM_004333) fusion: c.768:GTF2I_c.1177+1717:BRAFinv | Protein Fusion: mid-exon {GTF2I:BRAF}	NA
EVT-P-0000828-T05-IM6-135	P-0000828-T05-IM6		GTF2IRD1-BRAF	GTF2IRD1	in-frame	three_prime	140483546	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		GTF2IRD1 (NM_005685) - BRAF (NM_004333) rearrangement: c.2320+14147:GTF2IRD1_c.1178-589:BRAFinv | Protein Fusion: in frame {GTF2IRD1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0046745-T01-IM6-136	P-0046745-T01-IM6		KCND2-BRAF	KCND2	out-of-frame	three_prime	140485867	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KCND2 (NM_012281) - BRAF (NM_004333) fusion: c.1115+91797:KCND2_c.1177+1481:BRAFinv | Protein Fusion: out of frame {KCND2:BRAF}	NA
EVT-P-0056952-T01-IM6-138	P-0056952-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493821	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-141:KIAA1549_c.1140+287:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0044270-T02-IM6-139	P-0044270-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491645	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1560:KIAA1549_c.1140+2463:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0027592-T01-IM6-140	P-0027592-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492343	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5247+769:KIAA1549_c.1140+1765:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0055099-T03-IM6-141	P-0055099-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490663	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+1357:KIAA1549_c.1141-3279:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058584-T01-IM7-142	P-0058584-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491493	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+1663:KIAA1549_c.1140+2615:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066030-T01-IM7-143	P-0066030-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482785	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1172:KIAA1549_c.1314+36:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T01-IM6-144	P-0035153-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon15 fused with BRAF exon9) : c.4930-2046:KIAA1549_c.1140+3015:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035153-T03-IM6-145	P-0035153-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491093	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2046:KIAA1549_c.1140+3015:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0049925-T01-IM6-146	P-0049925-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489497	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+301:KIAA1549_c.1141-2113:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035844-T03-IM7-147	P-0035844-T03-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0035844-T04-IM7-148	P-0035844-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493285	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4033-1202:KIAA1549_c.1140+823:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T02-IM6-149	P-0017693-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-495:KIAA1549_c.1141-2153:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0062963-T01-IM7-150	P-0062963-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482198	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-27:KIAA1549_c.1314+623:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0012575-T01-IM5-151	P-0012575-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490295	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+793:KIAA1549_c.1141-2911:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052411-T05-IM7-152	P-0052411-T05-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0034117-T01-IM6-153	P-0034117-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140483812	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4346-57:KIAA1549_c.1178-855:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0052411-T03-IM6-154	P-0052411-T03-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493530	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4930-2296:KIAA1549_c.1140+578:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0009097-T01-IM5-155	P-0009097-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488817	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.5247+3837:KIAA1549_c.1141-1433:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0005121-T02-IM5-156	P-0005121-T02-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491248	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 with BRAF exons 9-18): c.4929+2454:KIAA1549_c.1140+2860:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0064872-T01-IM7-157	P-0064872-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489277	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4032+4347:KIAA1549_c.1141-1893:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0013777-T01-IM5-158	P-0013777-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481674	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549(NM_001164665) - BRAF (NM_004333) rearrangement: c.5248-2734:KIAA1549_c.1315-181:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0040760-T01-IM6-159	P-0040760-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140481540	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2627:KIAA1549_c.1315-47:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0014191-T01-IM6-160	P-0014191-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492679	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion : c.5248-320: KIAA1549_c.1140+1429:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0014381-T01-IM6-161	P-0014381-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140492888	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused to BRAF exon 9 -18): c.4930-2082:KIAA1549_c.1140+1220:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0028611-T01-IM6-162	P-0028611-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0015540-T01-IM6-163	P-0015540-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489119	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in frame with BRAF exons 9-18, which include the BRAF kinase domain): c.4930-69:KIAA1549_c.1141-1735:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA
EVT-P-0062070-T01-IM7-164	P-0062070-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490443	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-1064:KIAA1549_c.1141-3059:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0022404-T01-IM6-165	P-0022404-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482177	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused with BRAF exons 11-18) : c.5248-2270:KIAA1549_c.1314+644:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0004032-T01-IM5-166	P-0004032-T01-IM5		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487892	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exon 16 fused to BRAF exon 9) : c.5248-1949:KIAA1549_c.1141-508:BRAFdup | Protein fusion: in frame (KIAA1549-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043372-T01-IM6-167	P-0043372-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140487523	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.4929+2204:KIAA1549_c.1141-139:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0028611-T02-IM6-168	P-0028611-T02-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140482786	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 11-18): c.5248-4149:KIAA1549_c.1314+35:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0043874-T01-IM6-169	P-0043874-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490001	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-2907:KIAA1549_c.1141-2617:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0069024-T04-IM7-170	P-0069024-T04-IM7		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492117	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5059:KIAA1549_c.1140+1991:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0029361-T01-IM6-171	P-0029361-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489625	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion: c.4930-2667:KIAA1549_c.1141-2241:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016557-T01-IM6-172	P-0016557-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488576	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-15 fused in-frame to BRAF exons 9-18) : c.4929+2955:KIAA1549_c.1141-1192:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0032982-T01-IM6-173	P-0032982-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491506	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) Fusion (KIAA1549 exons 1-16 fused with BRAF exons 9-18) : c.5248-10:KIAA1549_c.1140+2602:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0064310-T04-IM7-174	P-0064310-T04-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140493446	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5247+3560:KIAA1549_c.1140+662:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0021127-T01-IM6-175	P-0021127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140491741	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-10 fused with BRAF exons 9-18) : c.4032+3996:KIAA1549_c.1140+2367:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0017693-T01-IM6-176	P-0017693-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140489537	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (c.1141-2153): c.4930-495:KIAA1549_c.1141-2153:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020125-T01-IM6-177	P-0020125-T01-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492175	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-17 fused to BRAF exons 9-18): c.5260:KIAA1549_c.1140+1933:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0045683-T02-IM6-178	P-0045683-T02-IM6		KIAA1549-BRAF	KIAA1549	unknown	three_prime	140492850	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5189:KIAA1549_c.1140+1258:BRAFdup | Protein Fusion: mid-exon {KIAA1549:BRAF}	NA
EVT-P-0019127-T01-IM6-179	P-0019127-T01-IM6		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140490657	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion (KIAA1549 exons 1-16 fused to BRAF exons 9-18): c.5247+2110:KIAA1549_c.1141-3273:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0066022-T01-IM7-180	P-0066022-T01-IM7		KIAA1549-BRAF	KIAA1549	in-frame	three_prime	140488035	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KIAA1549 (NM_001164665) - BRAF (NM_004333) fusion: c.5248-4074:KIAA1549_c.1141-651:BRAFdup | Protein Fusion: in frame {KIAA1549:BRAF}	NA
EVT-P-0036680-T01-IM6-181	P-0036680-T01-IM6		KLHL12-BRAF	KLHL12	in-frame	three_prime	140493984	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		KLHL12 (NM_021633) - BRAF (NM_004333) fusion (KLHL12 exons 1-4 fused to BRAF exons 9-18): t(1;7)(q32.1;q34)(chr1:g.202880970::chr7:g.140493984) | Protein Fusion: in frame {KLHL12:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047833-T02-IM6-183	P-0047833-T02-IM6		LRBA-BRAF	LRBA	in-frame	three_prime	140493444	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		LRBA (NM_001199282) - BRAF (NM_004333) fusion: t(4;7)(q31.3;q34)(chr4:g.151502269::chr7:g.140493444) | Protein Fusion: in frame {LRBA:BRAF}	NA
EVT-P-0006412-T01-IM5-184	P-0006412-T01-IM5		LUC7L2-BRAF	LUC7L2	unknown	five_prime	140501109	6	287	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		LUC7L2 (NM_001244584) - BRAF (NM_004333) Rearrangement : c.1008-2090:LUC7L2_c.860+103:BRAFdel | Antisense fusion	NA
EVT-P-0026158-T04-IM6-185	P-0026158-T04-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) fusion: c.1417-9863:MAD1L1_c.1140+2359:BRAFdup | Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0026158-T01-IM6-186	P-0026158-T01-IM6		MAD1L1-BRAF	MAD1L1	in-frame	three_prime	140491749	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MAD1L1 (NM_001013837) - BRAF (NM_004333) Rearrangement : c.1417-9863_c.1140+2359dup | Protein Fusion: in frame {MAD1L1:BRAF}	NA
EVT-P-0006601-T03-IM6-187	P-0006601-T03-IM6		MINDY4-BRAF	MINDY4	in-frame	three_prime	140490696	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		FAM188B (NM_032222) - BRAF(NM_004333) Fusion (FAM188B exon5 fused with BRAF exon9) : c.664-366:FAM188B_c.1141-3312:BRAF | Protein Fusion: in frame {FAM188B:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023220-T01-IM6-188	P-0023220-T01-IM6		MKLN1-BRAF	MKLN1	in-frame	three_prime	140496796	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKLN1 (NM_013255) - BRAF (NM_004333) rearrangement: c.98+4070: MKLN1_c.981-2529:BRAFinv | Protein Fusion: in frame {MKLN1:BRAF}	NA
EVT-P-0035393-T02-IM6-189	P-0035393-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140488436	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.771+833:MKRN1_c.1141-1052:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0009024-T01-IM5-190	P-0009024-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.772-184G: MKRN1_c.1140+623:BRAF dup | Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0008083-T03-IM6-191	P-0008083-T03-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484511	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.315-432: MKRN1_c.1178-1554:BRAF dup | Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-0001036-T01-IM3-192	P-0001036-T01-IM3		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482250	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NA | Protein fusion: in frame (MKRN1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047135-T01-IM6-193	P-0047135-T01-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140492510	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-972:MKRN1_c.1140+1598:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0004373-T02-IM5-194	P-0004373-T02-IM5		MKRN1-BRAF	MKRN1	unknown	three_prime	140434466	18	744	False	lost	0.0	False		Protein kinase domain; RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.314+4351:MKRN1_c.2232:BRAFdup | Protein fusion: mid-exon (MKRN1-BRAF)	NA
EVT-P-0009024-T02-IM6-195	P-0009024-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140493485	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion: c.772-185:MKRN1_c.1140+623:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA
EVT-P-0031896-T02-IM6-196	P-0031896-T02-IM6		MKRN1-BRAF	MKRN1	in-frame	three_prime	140482581	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-4 fused to BRAF exons 11-18): c.772-1018:MKRN1_c.1314+240:BRAFdup | Protein Fusion: in frame {MKRN1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008083-T02-IM6-197	P-0008083-T02-IM6		MKRN1-BRAF	MKRN1	out-of-frame	three_prime	140484505	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) fusion (MKRN1 exons 1-2 fused with BRAF exons 10-18): c.315-428:MKRN1_c.1178-1548:BRAFdup | Protein Fusion: out of frame {MKRN1:BRAF}	NA
EVT-P-0013536-T01-IM5-198	P-0013536-T01-IM5		MKRN1-BRAF	MKRN1	in-frame	three_prime	140483618	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		MKRN1 (NM_013446) - BRAF (NM_004333) rearrangement: c.544+172:MKRN1_c.1178-661:BRAFdup | Protein fusion: in frame (MKRN1-BRAF)	NA
EVT-P-0048036-T01-IM6-199	P-0048036-T01-IM6		NEO1-BRAF	NEO1	in-frame	three_prime	140497651	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NEO1 (NM_002499) - BRAF (NM_004333) fusion: t(7;15)(q34;q24.1)(chr7:g.140497651::chr15:g.73575682) | Protein Fusion: in frame {NEO1:BRAF}	NA
EVT-P-0043972-T03-IM6-200	P-0043972-T03-IM6		NRF1-BRAF	NRF1	in-frame	three_prime	140487712	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.607-7410:NRF1_c.1141-328:BRAFinv | Protein Fusion: in frame {NRF1:BRAF}	NA
EVT-P-0065446-T01-IM7-201	P-0065446-T01-IM7		NRF1-BRAF	NRF1	in-frame	three_prime	140483810	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		NRF1 (NM_005011) - BRAF (NM_004333) fusion: c.1349-10893:NRF1_c.1178-853:BRAFinv | Protein Fusion: in frame {NRF1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059002-T01-IM7-202	P-0059002-T01-IM7		OSBPL7-BRAF	OSBPL7	in-frame	three_prime	140491530	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		OSBPL7 (NM_145798) - BRAF (NM_004333) fusion: t(7;17)(q34;q21.32)(chr7:g.140491530::chr17:g.45890381) | Protein Fusion: in frame {OSBPL7:BRAF}	NA
EVT-P-0054958-T02-IM6-203	P-0054958-T02-IM6		PAK2-BRAF	PAK2	in-frame	three_prime	140486065	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PAK2 (NM_002577) - BRAF (NM_004333) rearrangement: t(3;7)(q29;q34)(chr3:g.196530579::chr7:g.140486065) | Protein Fusion: in frame {PAK2:BRAF}	NA
EVT-P-0008789-T01-IM5-205	P-0008789-T01-IM5		PHTF2-BRAF	PHTF2	in-frame	three_prime	140489236	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PHTF2 (NM_001127358) - BRAF (NM_004333) rearrangement: c.2224-367:PHTF2_c.1141-1852:BRAFinv | Protein fusion: in frame (PHTF2-BRAF)	NA
EVT-P-0044819-T01-IM6-206	P-0044819-T01-IM6		PLPP1-BRAF	PLPP1	out-of-frame	three_prime	140482123	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PPAP2A (NM_176895) - BRAF (NM_004333) fusion: t(5;7)(q11.2;q34)(chr5:g.54789192::chr7:g.140482123) | Protein Fusion: out of frame {PPAP2A:BRAF}	NA
EVT-P-0009694-T01-IM5-207	P-0009694-T01-IM5		PRKAR1B-BRAF	PRKAR1B	in-frame	three_prime	140491855	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR1B (NM_001164758) -BRAF (NM_004333) Rearrangement : c.892-2551:PRKAR1B_c.1140+2253:BRAFdup | Protein fusion: in frame (PRKAR1B-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008567-T01-IM5-208	P-0008567-T01-IM5		PRKAR2B-BRAF	PRKAR2B	in-frame	three_prime	140486416	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PRKAR2B (NM_002736) - BRAF (NM_004333) rearrangement: c.307+8063:PRKAR2B_c.1177+932:BRAFinv | Protein fusion: in frame (PRKAR2B-BRAF)	NA
EVT-P-0041181-T01-IM6-209	P-0041181-T01-IM6		PTPRZ1-BRAF	PTPRZ1	in-frame	three_prime	140492428	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PTPRZ1 (NM_002851) - BRAF (NM_004333) fusion: c.5367+457:PTPRZ1_c.1140+1680:BRAFinv | Protein Fusion: in frame {PTPRZ1:BRAF}	NA
EVT-P-0012884-T01-IM5-210	P-0012884-T01-IM5		PWWP2A-BRAF	PWWP2A	in-frame	three_prime	140495243	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		PWWP2A (NM_001130864) - BRAF(NM_004333) Rearrangement : t(5;7)(q33.3;q34)(chr5:g.159539324::chr7:g.140495243) | Protein fusion: in frame (PWWP2A-BRAF)	NA
EVT-P-0068260-T01-IM7-211	P-0068260-T01-IM7		RBMS3-BRAF	RBMS3	in-frame	three_prime	140482707	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RBMS3 (NM_001003793) - BRAF (NM_004333) fusion: t(3;7)(p24.1;q34)(chr3:g.30030473::chr7:g.140482707) | Protein Fusion: in frame {RBMS3:BRAF}	NA
EVT-P-0064764-T01-IM7-212	P-0064764-T01-IM7		RGS3-BRAF	RGS3	out-of-frame	three_prime	140493282	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RGS3 (NM_144488) - BRAF (NM_004333) fusion: t(7;9)(q34;q32)(chr7:g.140493282::chr9:g.116291333) | Protein Fusion: out of frame {RGS3:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041308-T02-IM6-213	P-0041308-T02-IM6		RRBP1-BRAF	RRBP1	in-frame	three_prime	140485917	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		RRBP1 (NM_004587) - BRAF (NM_004333) fusion: t(7;20)(q34;p12.1)(chr7:g.140485917::chr20:g.17634546) | Protein Fusion: in frame {RRBP1:BRAF}	NA
EVT-P-0007392-T01-IM5-214	P-0007392-T01-IM5		SCRIB-BRAF	SCRIB	in-frame	three_prime	140482150	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SCRIB (NM_182706) - BRAF (NM_004333) rearrangement: t(7;8)(q34;q24.3)(chr7:g.140482150::chr8:g.144876675) | Protein fusion: in frame (SCRIB-BRAF)	NA
EVT-P-0056639-T01-IM6-215	P-0056639-T01-IM6		SETBP1-BRAF	SETBP1	in-frame	three_prime	140486594	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SETBP1 (NM_015559) - BRAF (NM_004333) fusion: t(7;18)(q34;q12.3)(chr7:g.140486594::chr18:g.42638299) | Protein Fusion: in frame {SETBP1:BRAF}	NA
EVT-P-0048950-T01-IM6-216	P-0048950-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140492033	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+30037:SND1_c.1140+2075:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0012395-T01-IM5-217	P-0012395-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140493527	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1152+27174:SND1_c.1140+581:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0009619-T01-IM5-218	P-0009619-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482494	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1968+2394:SND1_c.1314+327:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0052299-T04-IM6-219	P-0052299-T04-IM6		SND1-BRAF	SND1	unknown	three_prime	140482935	10	400	False	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1038+2929:SND1_c.1200:BRAFinv | Protein Fusion: mid-exon {SND1:BRAF}	NA
EVT-P-0001687-T01-IM3-220	P-0001687-T01-IM3		SND1-BRAF	SND1	in-frame	three_prime	140490224	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) inversion: c.1152+28381_c.1141-2840inv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0004103-T01-IM5-221	P-0004103-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140481685	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1038+5799:SND1_c.1315-192:BRAFinv. | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0029727-T01-IM6-222	P-0029727-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140482257	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+12095:SND1_c.1314+564:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0068194-T01-IM7-223	P-0068194-T01-IM7		SND1-BRAF	SND1	in-frame	three_prime	140489175	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion: c.1152+17686:SND1_c.1141-1791:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0008140-T01-IM5-224	P-0008140-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489439	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-9 fused with BRAF exons 9-18) : c.1039-3849:SND1_c.1141-2055:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0034168-T01-IM6-225	P-0034168-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489421	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused to BRAF exons 9-18): c.1152+6972:SND1_c.1141-2037:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0025075-T01-IM6-226	P-0025075-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140490546	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1- 9 with BRAF exons 9-18) : c.1039-2471:SND1_c.1141-3162:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0002071-T02-IM5-227	P-0002071-T02-IM5		SND1-BRAF	SND1	in-frame	three_prime	140489292	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 fused with BRAF exons 9-18) : c.1152+18606:SND1_c.1141-1908:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0002197-T01-IM3-228	P-0002197-T01-IM3		SND1-BRAF	SND1	in-frame	five_prime	140488303	8	380	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		c.1528-8436:SND1_c.1141-922:BRAFinv | Protein fusion: in frame (BRAF-SND1)	NA
EVT-P-0007533-T01-IM5-229	P-0007533-T01-IM5		SND1-BRAF	SND1	in-frame	three_prime	140482226	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) fusion (SND1 exons 1-10 with BRAF exons 11-18) : c.1152+31738:SND1_c.1314+595:BRAFinv | Protein fusion: in frame (SND1-BRAF)	NA
EVT-P-0036567-T01-IM6-230	P-0036567-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140493552	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) Fusion (SND1 exon10 fused with BRAF exon9) : c.1153-22013:SND1_ c.1140+556:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0034845-T01-IM6-231	P-0034845-T01-IM6		SND1-BRAF	SND1	in-frame	three_prime	140489241	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SND1 (NM_014390) - BRAF (NM_004333) rearrangement: c.1527+3658:SND1_c.1141-1857:BRAFinv | Protein Fusion: in frame {SND1:BRAF}	NA
EVT-P-0054364-T01-IM6-232	P-0054364-T01-IM6		SNX8-BRAF	SNX8	in-frame	three_prime	140488730	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		SNX8 (NM_013321) - BRAF (NM_004333) fusion: c.1284+565:SNX8_c.1141-1346:BRAFdup | Protein Fusion: in frame {SNX8:BRAF}	NA
EVT-P-0026773-T01-IM6-234	P-0026773-T01-IM6		SVOPL-BRAF	SVOPL	unknown	five_prime	140482097	10	438	True	lost	0.0	False	RAS-binding domain; Cysteine-rich domain	Protein kinase domain		BRAF (NM_004333) rearrangement: c.1315-604:BRAF_chr7:g.138277426del | Transcript Fusion {BRAF:SVOPL}	NA

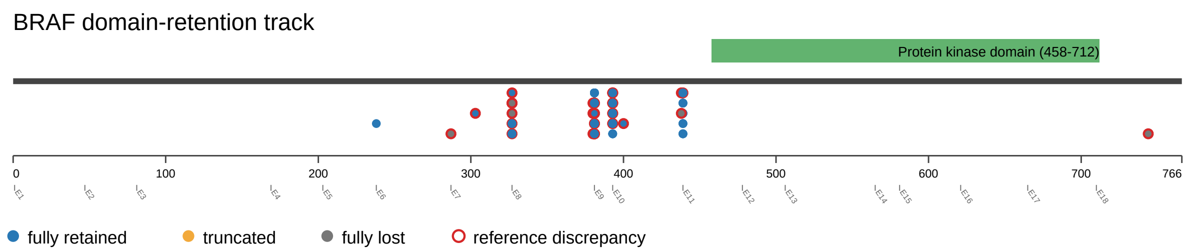
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051214-T01-IM6-235	P-0051214-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140502592	6	238	True	retained	1.0	False	Protein kinase domain	RAS-binding domain	Cysteine-rich domain	TMPRSS2 (NM_001135099) - BRAF (NM_004333) fusion: t(7;21)(q34;q22.3)(chr7:g.140502592::chr21:g.42867846) | Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-0018524-T01-IM6-236	P-0018524-T01-IM6		TMPRSS2-BRAF	TMPRSS2	out-of-frame	three_prime	140490337	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(21;7)(q22.3;q34)(chr21:g.42873801::chr7:g.140490337) | Protein Fusion: out of frame {TMPRSS2:BRAF}	NA
EVT-P-0022778-T01-IM6-237	P-0022778-T01-IM6		TMPRSS2-BRAF	TMPRSS2	in-frame	three_prime	140483147	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TMPRSS2 (NM_001135099) - BRAF (NM_004333) Rearrangement : t(7;21)(q34;q22.3)(chr7:g.140483147::chr21:g.42878796) | Protein Fusion: in frame {TMPRSS2:BRAF}	NA
EVT-P-0043836-T01-IM6-239	P-0043836-T01-IM6		TRA2B-BRAF	TRA2B	in-frame	three_prime	140491978	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRA2B (NM_004593) - BRAF (NM_004333) fusion: t(3;7)(q27.2;q34)(chr3:g.185650761::chr7:g.140491978) | Protein Fusion: in frame {TRA2B:BRAF}	NA
EVT-P-0054430-T02-IM6-240	P-0054430-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487875	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2719-489:TRIM24_c.1141-491:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025699-T02-IM6-241	P-0025699-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140486854	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2014+674:TRIM24_c.1177+494:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0045111-T01-IM6-242	P-0045111-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140488049	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2257-760:TRIM24_c.1141-665:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0066835-T01-IM7-243	P-0066835-T01-IM7		TRIM24-BRAF	TRIM24	in-frame	three_prime	140490589	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1879-502:TRIM24_c.1141-3205:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0068451-T01-IM7-244	P-0068451-T01-IM7		TRIM24-BRAF	TRIM24	in-frame	three_prime	140482765	11	439	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.484-860:TRIM24_c.1314+56:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0030988-T01-IM6-245	P-0030988-T01-IM6		TRIM24-BRAF	TRIM24	unknown	three_prime	140491002	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.2349:TRIM24_c.1140+3106:BRAFinv | Protein Fusion: mid-exon {TRIM24:BRAF}	NA

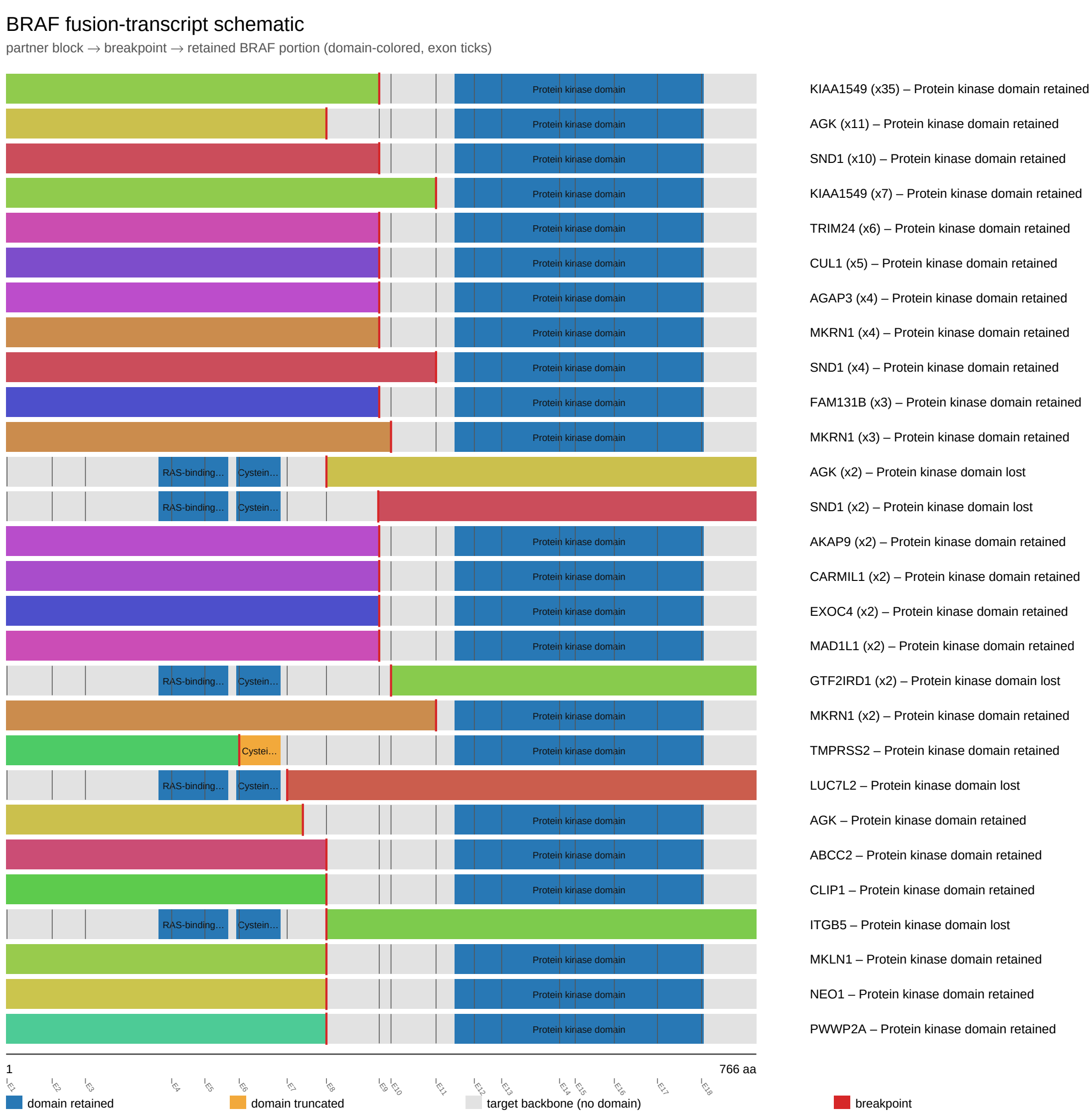
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032569-T01-IM6-246	P-0032569-T01-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) Rearrangement : c.1878+791:TRIM24_c.1141-552inv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0032569-T02-IM6-247	P-0032569-T02-IM6		TRIM24-BRAF	TRIM24	in-frame	three_prime	140487936	9	381	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		TRIM24 (NM_015905) - BRAF (NM_004333) fusion: c.1878+791:TRIM24_c.1141-552:BRAFinv | Protein Fusion: in frame {TRIM24:BRAF}	NA
EVT-P-0019039-T01-IM6-248	P-0019039-T01-IM6		UBE2H-BRAF	UBE2H	out-of-frame	three_prime	140499248	8	327	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		BRAF (NM_004333) - UBE2H (NM_003344) rearrangement: c.980+914:BRAF_c.299-4611:UBE2Hdup | Protein Fusion: out of frame {UBE2H:BRAF}	NA
EVT-P-0027762-T01-IM6-250	P-0027762-T01-IM6		ZC3HAV1-BRAF	ZC3HAV1	in-frame	three_prime	140485946	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZC3HAV1 (NM_020119) - BRAF (NM_004333) rearrangement: c.698-431:ZC3HAV1_c.1177+1402:BRAFdup | Protein Fusion: in frame {ZC3HAV1:BRAF}	NA
EVT-P-0008920-T01-IM5-251	P-0008920-T01-IM5		ZNF207-BRAF	ZNF207	in-frame	three_prime	140485847	10	393	True	retained	1.0	False	Protein kinase domain	RAS-binding domain; Cysteine-rich domain		ZNF207 (NM_001098507) - BRAF (NM_004333) rearrangement: t(7;17)(q34;q11.2)(chr7:g.140485847::chr17:g.30687533) | Protein fusion: in frame (ZNF207-BRAF)	NA

Figures

domain retention outliers

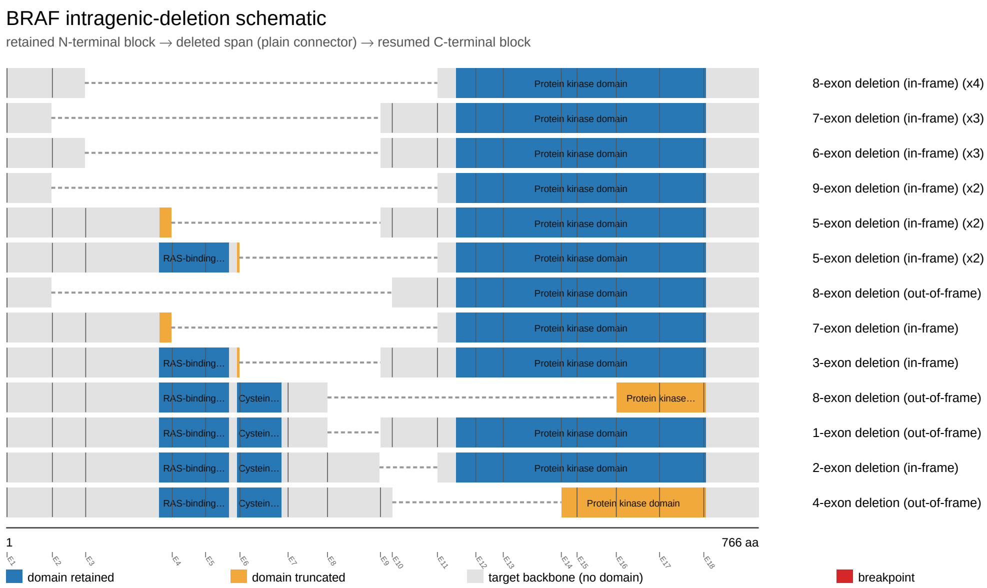


fusion schematic



Showing the top 28 of 89 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison